

RESEARCH ARTICLE

Real-Time Athlete Fatigue Monitoring Using Fuzzy Decision Support Systems

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Abstract

Sports scientists worry about fatigue, because it affects performance, increases injury risk, and harms health. Traditional fatigue measurements may miss this complicated and ever-changing state, placing athletes at risk of injury while training. This study tests the premise that cutting-edge, real-time monitoring devices improve athlete health and performance. This research aims to reduce tiredness by developing a Fuzzy Decision Support System for Real-Time Athlete Weariness Monitoring. Fuzzy logic can handle unclear performance data, making it a more flexible and advanced alternative to standard methods. Sports athlete fatigue is complicated and dynamic, requiring improved, more precise, and real-time monitoring approaches. The FDSS-RAFM model uses fuzzy logic to account for human performance and physiology. The FDSS-RAFM model assesses athlete fatigue in a comprehensive and context-aware manner. This study's findings can help coaches, players, and sports scientists improve training programs, reduce injury risk, and improve performance in ever-changing athletic contexts. Fuzzy decision-support systems and other cutting-edge technology can improve athletes' health and performance, adding to sports science literature. Experimental results show that the proposed FDSS-RAFM model outperforms competing models in sensitivity (97%), specificity (89%), accuracy (96%), and dynamic adaptation error analysis (2.41%).

Keywords Athlete fatigue · Sports science · Fuzzy logic · Decision support systems · Injury risk · Physiological monitoring · Real-time assessment

1 Introduction

Sports science is concerned about fading energy, because it affects athletes' safety, performance, and health [1]. Due to fatigue's intricacy and dynamic character, which is affected by many psychological and physiological factors, intelligent and adaptable monitoring is needed [2]. Despite their prevalence, traditional monitoring methods cannot detect minor tiredness indicators [3]. This limits athletes' health and makes training less effective [4]. Current monitoring methods are inadequate, and new real-time technologies to control athlete weariness are needed immediately. This inspired our work [5]. New approaches are needed because of worrying effects like lower athletic performance and higher damage susceptibility [6]. Thus, our research aims to improve our understanding of fuzzy decision-support systems for real-time athlete weariness tracking.

Conventional machine learning models like decision trees, neural networks, and support vector machines (SVMs) need large, high-quality datasets to handle missing or ambiguous data, overcome hurdles, etc. [7]. The fuzzy decision-support system described for real-time physiological signal monitoring handles confusing or



erroneous data well. Sensor measurements may be imprecise in a dynamic and unexpected context like a sports arena due to perspiration, motion artifacts, or sudden activity changes, making machine learning models less effective. Fuzzy logic helps the system detect fatigue more precisely and interpretably, even with noisy or inaccurate data. This shows its value and efficacy compared to older machine learning methods [8]. Fatigue, including moderate-to-severe effort-related impairment, affects athletes' performance. All these states are fatigued for this study, and weariness and exhaustion are used for moderate and severe fatigue. Real-time tiredness tracking may help athletes manage performance and avoid injury [9].

The unique solution uses fuzzy logic to manage human performance data's inherent uncertainties and imprecisions [10]. Fuzzy logic may work in ambiguous situations instead of linear models and black-and-white categories [11]. This new knowledge allows a more flexible approach to athlete fatigue, which is complicated and ever-changing [12]. The intricacy and ever-changing nature of athlete weariness in current sports make conventional monitoring technologies unsuitable [13]. Instead of using outmoded approaches, releasing a new age of monitoring that provides real-time insights is tough [14]. This project aims to fill a gap in the literature by developing and using a fuzzy decision-support system to assess athletes' weariness.

Fuzzy decision-support systems offer an alternative to binary frameworks in decision-making models. Fuzzy logic [15] handles imprecision and ambiguity better and is context-aware. This study uses fuzzy decision-support systems to evaluate athlete weariness and provide a fuller picture. The goal is to document the complex relationship between physiological and performance indicators. The academic community and society at large value this discovery. In the ever-changing world of professional sports, coaches, players, and experts may always improve performance, prevent injuries, and maximize training [16]. Fuzzy decision-support systems for athlete monitoring strengthen sports science theory [17], demonstrating the practical use of new technology [18].

In conclusion, the FDSS-RAFM model advances sports science and offers a new perspective on athlete weariness. The research aims to show how fuzzy decision-support systems might help athletes manage weariness by offering real-time insight into its complex and ever-changing nature, enabling a more informed and flexible strategy for their well-being and performance [19]. This study recommends using fuzzy decision-support systems and other technology to detect and control athlete fatigue. It proposes a major paradigm shift in this sector. Its impact on sports science and the athletes it aims to inspire is huge.

This study significantly contributes by developing FDSS-RAFM,

- (i) A real-time monitoring tool for athlete fatigue enables coaches to make better judgments during training and competition and reduces performance declines.
- (ii) To emphasize the need for holistic athlete care and fuzzy decision-support systems in sports science to improve athlete monitoring.
- (iii) Show how fuzzy decision-support systems can improve performance, prevent injuries, and maximize training in the ever-changing sports world.
- (iv) Understanding athlete weariness, implementing fuzzy logic into decision-support systems, and encouraging groundbreaking sports science computational research.

The paper's outline continues: Sect. 2 reviews contemporary sports science and athlete monitoring literature. The suggested FDSS-RAFM model's theoretical basis and components are explained in Sect. 3. Experiment execution and analysis for the proposed study are described in Sect. 4. Section 5 summarizes the paper's findings.

2 Related Works

Ryan et al. [20] evaluated athlete monitoring systems in Australian football (AF) to determine their effects on training programs. It evaluates preparation based on training load and individual tolerance, analyzing existing research and empirical studies. The data sources are sports science databases, peer-reviewed publications, and

conference proceedings. The results present a complete study of theoretical frameworks and their practical application in AF contexts and a list of topics for further empirical research. The study will raise concerns about the lack of theoretical framework research and whether monitoring systems are over- or under-reliant on exogenous shocks.

Neupert et al. [21] evaluated athlete monitoring system data collection, processing, and distribution in elite UK sports. Survey-based AMS procedures involve participants, data collecting, and analysis. Discoveries will attempt to ascertain the following: how widespread AMS is, which monitoring techniques are most often employed, the degree to which ASRM questionnaires exhibit variability, and how efficient feedback procedures are. Among the many areas where the study finds AMS to be lacking are the following: questionnaire standardization, improved feedback processes, integrating AMS data into coaching practices, and achieving harmony between research findings and real application.

The Mobile Athlete Self-Report Measure (M-ASRM) was used in semi-structured interviews by Duignan et al. [22] to investigate how stakeholders in top Gaelic Games perceived it. Learn everything there is to know about stakeholder perceptions: From purpose clarity and implementation techniques to players' impressions of use and facilitators of M-ASRM use: From the results. The study's overarching goal is to fill knowledge gaps in stakeholder misunderstandings and underestimating practical requirements for effective implementation. Among the implementation suggestions are filling in the blanks regarding shared decision-making, educational tactics, feedback avenues, and the practicality and accountability of planning.

Injury risk in elite male youth footballers was investigated by Johnson et al. [23] about exposure time and growth-related variables. Analysis of growth characteristics, training volume, exposure duration, and injury monitoring was carried out with the help of 49 participants. The findings revealed four relationships: one between growth rate and injury incidence (a linear relationship), one between lower limb growth rate and injury incidence (a non-linear relationship), one between injury burden and percentage of expected adult stature (a non-linear relationship), and a linear association between lower limb growth rate and injury incidence overall. Additionally, the study found that exposure time varied from week to week and that there were peak risk times. The results highlight the importance of additional studies addressing growth variables, interaction effects, individual variability, longitudinal data, and intervention tactics to reduce the risk of injury in young football players.

Professional volleyball players are at increased risk for overuse injuries, and de Leeuw et al. [24] suggested a machine learning study to find signs of training load and wellness that can foretell these problems. During the 2018 international season, data were collected from 14 top players. The study's overarching goals are to (1) identify individual differences in injury risk and (2) highlight the significance of jump load as a predictive factor. The results also point to potential avenues for future research, including longitudinal studies, examinations of exogenous variables, validation of prediction models, explorations of intervention tactics, and incorporation of individualized monitoring data into injury prevention initiatives.

The severity of injuries substantially impacted the subjective well-being of 386 elite teenage athletes in a study proposed by Rosen and Heijne [25]. There was an increased incidence of reported severe injuries among those with lower well-being ratings. Injuries occurred 5.6% less frequently for those with higher well-being scores. Less severe injuries were connected with higher well-being scores. Concerning topics, such as the effects of well-being on injury outcomes over the long term, psychosocial factors, therapies to increase well-being, contextual factors, and the incorporation into injury management protocols, the study seeks to identify gaps in current knowledge and proposes opportunities for future research.

Improta et al. [26] assessed renal function in post-transplant patients using data from the University Hospital Federico II of Naples' Department of Nephrology through a Clinical Decision Support System (CDSS) based on Fuzzy Logic. Fuzzy sets and rules informed by clinical data and expert knowledge will enable the system to achieve an accuracy rate of over 90%. This technology will also assess the clinical usefulness and visibility of patient status. Findings from this study will help fill in some of the gaps in understanding topics, including medication error impact, long-term evaluation, integration with EHRs, generalizability across patient demographics, physician feedback, and continuous learning.

A Fuzzy Decision Support System for the Diagnosis of Coronary Artery Disease Utilizing Datasets from the University of California Irvine (UCI) was the goal of Setiawan et al. [27]. The system extracts rules using Rough Set Theory and then fuzzifies them to make diagnostics more accurate. Results from the Ipoh Specialist Hospital and the University of California, Irvine's heart disease databases confirmed the system's efficacy. Compared to cardiologists and angiography, its percentage estimations of coronary artery obstruction are more accurate. Additional areas that could use further investigation are the following: user-friendliness, cross-cultural concerns, rule explainability, user-friendliness, clinical decision impact, multi-modal data integration, and long-term validation.

Sloane and Silva [28] investigated healthcare AI's potential for rule-based expert systems (ES) and machine learning (ML) integration. Clinical decision-support systems using fuzzy logic and algorithms are the primary focus of this article. This research shows that AI categories are very flexible, that fuzzy logic is excellent at dealing with confusing data, and that various algorithms can provide accurate diagnoses and decision assistance. As an example of effective AI applications in healthcare, it highlights possible areas for future study that lack sufficient coverage, such as human–AI collaboration, ethics, interoperability, the ability to explain AI judgments, adaptability of systems over time, and tools that are easy to use.

Biró et al. [29] suggested the AI-Assisted Fatigue and Stamina Control for Performance Sports on IMU-Generated Multivariate Times Series Datasets. The 19 athletes were monitored throughout sessions using inertial measurement units (IMUs) for triaxial acceleration, angular velocity, and magnetic orientation. The standardized training program included rest periods and progressively more strenuous workouts. An ML model trained on the raw time series data was based on time-domain and frequency-domain properties. A comparison was made between the performances of the Random Forest, Gradient Boosting Machines, and LSTM networks. Prediction error and bias estimates were used via a feedback loop to make real-time adjustments to the model. Predicted degrees of exhaustion were significantly correlated with actual drops in performance, demonstrating the AI model's strong prediction accuracy for weariness. Stamina projections made Individualized training modifications possible, in harmony with the physiological limits of the athletes. Reduced systematic prediction errors were achieved via the use of bias correction methods. Additionally, the model was adjusted in real time, which improved training periodization tactics, decreased the likelihood of overtraining, and generally improved athletic performance.

Gan et al. [30] proposed the heart rate variability analysis method for exercise-induced fatigue monitoring. The research achieves a non-linear analysis of HRV by building dimensional and entropic characteristics on top of two-dimensional space. This is achieved by transforming a one-dimensional time series into two dimensions by a sequence of rotations and translations, quantifying the system's degree of chaos. Tests were conducted to assess the efficacy of the suggested method. Sixteen healthy adults participated in the study by doing one conventional resistance training session and two pressured Smith squats. Utilizing a support vector machine (SVM) classification model, this approach attained a classification accuracy of 0.829. Table 1 shows the summary of existing models.

The literature review highlights the relevance of healthcare machine learning, rule-based expert systems, clinical decision-support systems, injury risk assessment, and athlete monitoring. Personalized risk profiles, standardization, empirical testing, and AI adaptability are emphasized. Incorporating empirical testing, addressing well-being in monitoring, integrating fuzzy logic and machine learning algorithms for nuanced decision-making, considering long-term evaluation, and collaborating with stakeholders are all essential components of the proposed FDSS-RAFM strategy. Incorporating subjective well-being assessments into fatigue monitoring, emphasizing empirical testing, considering long-term evaluation, user-friendliness, and continual learning are all critical aspects of this system. Involving stakeholders throughout development and implementation helps clear any confusion and guarantees a smooth rollout. Incorporating empirical data and drawing on the varied insights offered by the literature review, the system should emphasize the health and performance of athletes in real-world settings.

Table 1 Summary of the related works

Author(s)	Method	Advantages	Limitations
Ryan et al. [20]	Review of theoretical frameworks and practical impact of athlete monitoring in AF	Comprehensive review of frameworks; insights into professional AF training practices	Lack of empirical testing of frameworks; underrepresentation of external stresses
Neupert et al. [21]	Survey-based analysis of AMS processes in top-tier UK sports	Identifies common AMS methods; highlights feedback and participant selection processes	Variability in ASRM questionnaires; limited integration of AMS data into coaching practices
Duignan et al. [22]	Semi-structured interviews on M-ASRM use in Gaelic Games	Highlights stakeholder perceptions; identifies facilitators for M-ASRM implementation	Gaps in shared decision-making, feedback channels, and implementation strategies
Johnson et al. [23]	Analysis of growth variables, training volume, and injury risk in youth footballers	Identifies linear and non-linear injury-growth relationships; recognizes peak injury risk times	Limited focus on longitudinal data, intervention tactics, and individual variability
de Leeuw et al. [24]	Machine learning analysis of training load and wellness data in professional volleyball players	Highlights predictive value of jump load; identifies individual injury risks	Limited data sample (14 players); lacks validation of models and exploration of exogenous factors
Rosen and Heijne [25]	Correlation analysis between well-being scores and injury incidence in elite youth athletes	Establishes a link between well-being and injury severity; emphasizes psychosocial factors	Lack of long-term studies on well-being; minimal exploration of contextual factors
Improta et al. [26]	Fuzzy Logic-based CDSS for renal function assessment in post-transplant patients	High diagnostic accuracy; integration of expert knowledge in decision-making	Limited generalizability; lacks long-term validation and multi-modal integration with EHRs
Setiawan et al. [27]	Fuzzy Logic and Rough Set Theory for coronary artery disease diagnosis	Improved diagnostic accuracy compared to traditional methods; validation across datasets	User-friendliness, rule explainability, and cross-cultural applicability require further study
Sloane and Silva [28]	Integration of rule-based expert systems with machine learning in healthcare AI	Highlights the flexibility of AI systems; fuzzy logic manages ambiguous data effectively	Gaps in explainability, ethics, adaptability, and human-AI collaboration

3 Athlete Fatigue Monitoring Strategy with FDSS Framework

Monitoring athletes' fatigue levels is an essential component of sports science as this affects their well-being, protection, and performance. Introducing an advanced method for tracking athlete weariness, particularly one that an FDSS supports, signals a paradigm change in how athlete fatigue is evaluated and managed in real time. Traditional monitoring methods are inadequate when assessing athlete weariness, which is caused by a complex interaction of psychological and physiological elements. The strategy for athlete fatigue monitoring using FDSS emphasizes the need for real-time information, acknowledging that fatigue is a dynamic phenomenon. Its goals extend beyond just improving training programs; they also include reducing the likelihood of injuries and generally boosting the standard of sports performance. The novel approach to sports science is guided by the theoretical framework of the model, which lays out the underlying principles, concepts, and methodologies for Fuzzy Decision Support Systems for Real-Time Athlete Fatigue Monitoring. The model's ability to capture and explain the complex and ever-changing nature of athlete tiredness is primarily due to the interrelated pieces that comprise this theoretical framework.

Figure 1 depicts the proposed FDSS-RAFM architecture, a high-level model developed to improve the evaluation of athlete tiredness in real time. Achieving this goal is accomplished via its distinct components. Athletes can be subtly fatigued, and fuzzy rule sets are essential for capturing the complex interactions between variables. A granular evaluation of characteristics like heart rate and exhaustion is made possible by membership functions, which objectively evaluate the degree to which specific values belong to fuzzy sets. Uncertainty is essential in athlete-related data, and intuitionistic triangular fuzzy numbers improve adaptation by clearly describing it. Athletes and coaches benefit from the real-time monitoring paradigm's rapid evaluations and decision-making capabilities. Fuzzy inference lets the model learn from mistakes and adapt to real-time sports occurrences. Stakeholder feedback improves collaboration and relevance, while longitudinal analysis ensures efficacy. These pieces form an all-encompassing, customizable architecture for precise, real-time athlete fatigue monitoring.

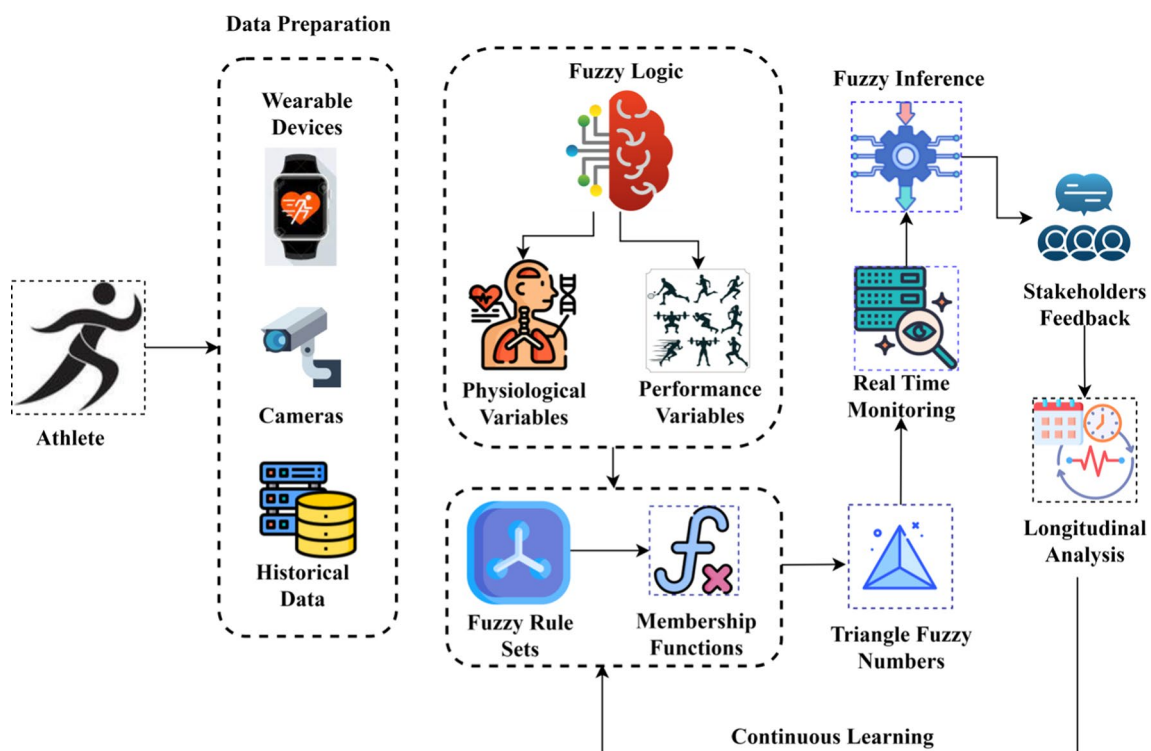


Fig. 1 System architecture of the FDSS-RAFM model

Figure 1 shows the FDSS-RAFM Model's system architecture tracks athletes' fatigue in real time using a detailed method. The first step, "Athlete Data Preparation", gathers data from records, wearables, and cameras. This diverse set covers vital physiological and performance fatigue metrics. Second, Fuzzy Logic is crucial for addressing human performance and physiology's intricacies. Next, Fuzzy Rule Sets and Membership Functions will rank fuzzy input variables by proper interpretation. Intuitionistic Triangle Fuzzy Numbers are used to capture athlete data ambiguity explicitly. Adding these vague components to Real-Time Monitoring can instantly assess athlete weariness. Fuzzy Inference adjusts the model to serve the athlete better. The feedback loops in Stakeholders keep everyone on the same page and consider sports experts, coaches, and players' opinions. The FDSS-RAFM Model was refined by iterative learning using Continuous Learning and Longitudinal Analysis data. A complicated and flexible system that uses real-time inputs, stakeholder interaction, and continuous development measures athlete exhaustion accurately. The entire design flow shows this system.

Smartwatches, fitness trackers, and security cameras can input health and fitness data. This dataset comprises heart rate, breathing, and physical activity measurements. These inputs examined by cutting-edge AI and ML models can better track and predict the user's health and physical activities. The approach uses a centralized data storage and analysis center with strong computational models and databases. Together, these components analyze and organize enormous data volumes for accurate, quick insights. It provides customized recommendations and detects anomalies and health risks using anomaly detection and predictive analytics. The output layer delivers fitness trends, health alarms, and suggested goals in an intelligible style. The FDSS-RAFM architecture uses real-time feedback loops to maximize health outcomes. It smoothly manages health monitoring and fitness tracking applications due to its scalability and agility.

3.1 FDSS: Unraveling the Intricacies of Athlete Fatigue

Sports science is evolving toward Fuzzy Decision Support Systems. Modern fuzzy logic-based FDSS frameworks underpin the unique method. Fuzzy logic, which considers more complicated aspects, better captures athlete weariness's dynamic and complex nature. This revolutionary improvement has made real-time athlete weariness measurement and management more flexible and sophisticated. A computer system must process complicated, unexpected, and incomplete data for FDSS to work. FDSS recognizes and compensates for ambiguity, making it ideal for the complicated domain of athlete weariness. Fuzzy decision-support systems (FDSS) use intuitionistic triangular fuzzy numbers, membership functions, and fuzzy rule sets to understand fitness-related attributes.

3.2 Dynamic Interaction of Physiological and Performance Variables

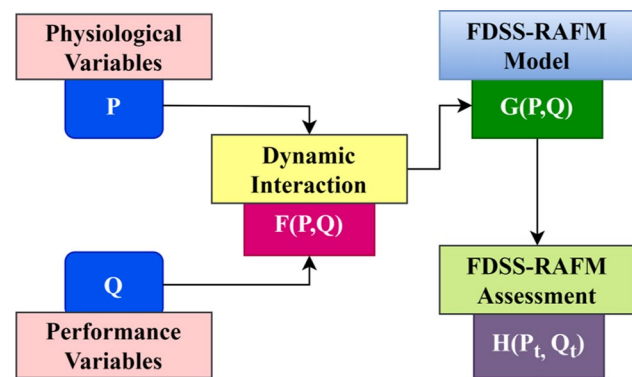
The theoretical framework emphasizes a holistic view of athlete fatigue within its dynamic environment. It explores its many facets and how they arise from the complex interaction of physiological and performance factors. Figure 2 depicts the complex interactions between many physiological and performance variables contributing to athlete fatigue. This recognition forms the theoretical basis of the FDSS-RAFM model.

Here, P stands for the physiological factors, and Q for the performance variables. Equation (1) can represent the dynamic interactions that lead to athlete fatigue as $F(P, Q)$

$$F(P, Q) \rightarrow \text{Multifaceted Athlete Fatigue}, \quad (1)$$

where F captures the complex relationship between the variables of performance (Q) and physiological state (P), and the FDSS-RAFM model is driven by this idea to take a comprehensive approach to its design and operation. Adding velocity, acceleration, core temperature, and heart rate to the model improves comprehension. This study does this to complete the model and avoid simplifying the aspects affecting athlete weariness. Since the model has many components (P and Q), Eq. (2) yields a formula.

$$\text{FDSS} - \text{RAFM Model} = G(P, Q). \quad (2)$$

Fig. 2 Dynamic interaction of physiological and performance variables

The model utilizes a complex and comprehensive technique, represented by variable G , to accurately assess athlete fatigue levels. This is done by considering their complex interdependencies. While the model is theoretically sound, its holistic approach has real-world ramifications. The FDSS-RAFM model accurately and quickly assesses athlete depletion by considering changing physiological and performance parameters. Advanced data improves the model's understanding of a player's status, allowing for more thorough and context-aware fatigue evaluation in real-time sports. In principle, the FDSS-RAFM model can accurately model athlete weariness. A detailed technique (G) enables accurate real-time assessments with real-world repercussions. The model captures athletes' situations more gently by including the dynamic nature of performance and physiological components (U and Q)

$$FDSS - RAFM \text{ Assessment} = H(P_t, Q_t). \quad (3)$$

In dynamic sports environments, the real-time assessment function (H in Eq. (3)) provides a comprehensive and context-sensitive evaluation of weariness status. It is affected by shifting dynamics, especially Pb and Qb . Since mathematical notations may express the theoretical framework, the model completely accounts for athlete depletion details.

3.3 Components of the FDSS Model for Athlete Fatigue Monitoring

The Fuzzy Decision Support System (FDSS) real-time athlete tiredness evaluation uses intuitionistic triangular fuzzy numbers, membership functions, and fuzzy rule sets. The system uses intuitionistic triangular fuzzy numbers and a membership function to assess the athlete's exhaustion and uncertainty.

3.3.1 Fuzzy Rule Sets

Athlete fatigue monitoring relies on fuzzy rule sets through Fuzzy Decision Support Systems. This function will help explain the intricate relationship between features and athlete weariness. This study discusses the applicability, adaptability, and importance of fuzzy rule sets in representing complicated variable interactions. FDSS outputs include athlete weariness evaluations based on fuzzy rule sets. These concepts strive to be more flexible than binary categories. This shift in focus may allow us to study athlete exhaustion's interconnections more thoroughly and creatively.

Traditional decision-making frameworks offer two options, but fuzzy rule sets offer several. Athletes know their data is erroneous and unclear and attempt to fix it. Fuzzy rule sets are adaptive and can accurately capture hazy attribute-level boundaries. Instead of labeling an athlete's exhaustion "high" or "low", flexible guidelines could account for varied intensities. This set of imprecise criteria cannot identify tiredness levels. Instead, they are adaptable to consider athlete weariness as a natural variable. Since an athlete's exhaustion might not necessarily fall into clear categories, the "fuzziness" makes for a more accurate portrayal.

In fuzzy rule sets for athlete fatigue monitoring, heart rate, sleep quality, soreness level, psychological condition, training load, and other input parameters should be considered. A low, moderate, or high heart rate is a physiological indication that shows the degree of physical activity. The quality of sleep influences levels of exhaustion and recovery. A high degree of stress can lead to fatigue, whereas a low level of soreness signals physical stress and the need for recovery. Training load refers to the total stress that training sessions inflict, and an athlete's emotional health might affect how they perceive exhaustion. Assume that $X_1, X_2, \dots, X_n$, are the input attributes and that Y is the output representing the athlete's weariness. Equation (4) provides an expression for a fuzzy rule set

$$\text{Rule}_i : \text{IF } X_1 \text{ is } A_{i1} \text{ AND } X_2 \text{ is } A_{i2} \text{ AND } \dots \text{ AND } X_n \text{ is } A_{in} \text{ THEN } Y \text{ is } B_i. \quad (4)$$

The linguistic term or membership function linked to the input X_j is represented by A_{ij} , while the linguistic term or function linked to output Y is represented by B_i .

Table 2 shows that in cases where numerous circumstances lead to elevated levels of weariness, Rules 1–4 reflect possible outcomes. When an athlete feels moderate exhaustion due to a balanced set of circumstances, rules 5–8 apply. In some instances, as Rules 9–12 indicate, several variables converge to reduce weariness. These guidelines lay the groundwork for the fuzzy logic system to evaluate and classify athlete exhaustion according to the input factors. The data's unique qualities and the system's intended granularity might enlarge or refine the rule set. Adjusting variable membership functions makes the fuzzy rule set more exact and versatile.

Fuzzy rule sets allow the FDSS to adapt and derive context-aware judgments, addressing athlete data's flaws and dynamics. A fuzzy rule set-adaptive monitoring system could evaluate athletes' exhaustion more accurately and faster. Because it uses multiple fuzzy rules to determine athlete exhaustion, the FDSS system is comprehensive. This method goes beyond simple categorizations, since athlete exhaustion is multi-dimensional. Due to their malleability, fuzzy rule sets can evolve. The system may adjust its rules based on new data to increase accuracy and efficiency. Fuzzy rule sets in FDSS use data to detect athlete exhaustion better than earlier techniques, because they are flexible. This cutting-edge method improves the system's ability to handle athlete-specific elements, improving understanding of sports science fatigue dynamics.

3.3.2 Membership Functions in Athlete Fatigue Monitoring

Any fuzzy decision-making system needs a membership function to determine how much a value belongs to a fuzzy set. This study analyzes membership functions to understand athlete fatigue better. FDSS relies on

Table 2 Fuzzy rule set for athlete fatigue monitoring

Rule No	Fuzzy rule
1	IF Heart Rate is High AND Sleep Quality is Poor, THEN Fatigue is High
2	IF Soreness Level is High AND Stress Level is High, THEN Fatigue is High
3	IF the Mood State is Negative AND the Previous Day's Performance is Below Average, THEN Fatigue is High
4	IF the Training Load is Heavy AND the Duration of the Training Session is Long, THEN Fatigue is High
5	IF Heart Rate is Moderate AND Sleep Quality is Good, THEN Fatigue is Moderate
6	IF the Soreness Level is Moderate AND the Stress Level is Moderate, THEN Fatigue is Moderate
7	IF the Mood State is Neutral AND the Previous Day's Performance is Average, THEN Fatigue is Moderate
8	IF the Training Load is Moderate AND the Duration of the Training Session is Moderate, THEN Fatigue is Moderate
9	IF Heart Rate is Low AND Sleep Quality is Good, THEN Fatigue is Low
10	IF Soreness Level is Low AND Stress Level is Low, THEN Fatigue is Low
11	IF the Mood State is Positive AND the Previous Day's Performance is Above Average, THEN Fatigue is Low
12	IF the Training Load is Light AND the Duration of the Training Session is Short, THEN Fatigue is Low

membership functions to better navigate athlete tiredness. Imagine X as the input property and A as a fuzzy set language phrase. The membership function $\mu_A(X)$ in fuzzy set A determines membership for a given X value. Equation (5) gives its expression

$$\mu_A(X) : X \rightarrow [0,1], \quad (5)$$

where X represents what fuzzy set percentage? The coefficient $\mu_A(X)$ represents the input attribute X and the fuzzy set A , with 0 indicating no membership and 1 total membership. Membership functions quantify a value's fuzzy set membership. Heart rate, sleep quality, pain, stress, and mood can be more precisely assessed in athlete fatigue monitoring. Since membership functions are continuous, FDSS can handle many circumstances. Because athlete data are ambiguous and imprecise, this saves the day. Membership functions assess the nuanced levels of several criteria rather than a binary classification. Equations (6)-(8) contain membership functions for each major athlete tiredness factor

$$\mu_{High\ Fatigue}(Fatigue) \quad (6)$$

$$\mu_{Low\ Sleep\ Quality}(Sleep\ Quality) \quad (7)$$

$$\mu_{Elevated\ Stress\ Level}(Stress\ Level). \quad (8)$$

Athlete weariness is unpredictable; therefore, processes assign different degrees of membership to attribute values. Since membership functions are malleable, FDSS can show more complex and realistic features down to the finest intergradations. FDSS membership features simplify athlete weariness monitoring and make it more precise and adaptable. FDSS's membership levels and continuum of choices have improved athlete fatigue knowledge and judgment in varied settings.

3.3.3 Intuitionistic Triangular Fuzzy Number

Integrating intuitionistic triangular fuzzy numbers greatly improves Fuzzy Decision Support Systems' adaptability and sophistication for athlete fatigue monitoring. This study uses the FDSS framework to study intuitionistic triangular fuzzy numbers' complexity and how they might be used to assess athlete tiredness. The intuitionistic triangular fuzzy number extends fuzzy set theory mathematically. Classical fuzzy numbers only include membership and non-membership, whereas intuitionistic triangular fuzzy numbers incorporate the degree of hesitation. This third dimension quantifies fuzzy value ambiguity or uncertainty. This expansion helps athlete fatigue monitoring data show its uncertainties and imprecisions. Let T be an intuitive triangular fuzzy number in Eq. (9)

$$T = (a, b, c; H). \quad (9)$$

The fuzzy set is represented by a on the left, b in the center, and c on the right. The uncertainty associated with membership value is represented by the degree of hesitation ($0 \leq H \leq 1$). Heart rate, sleep quality, discomfort, stress, and mood are likely ambiguous for athletes. Intelligent triangular fuzzy numbers actively quantify athlete tiredness ambiguity to better convey it. Combine the degree of hesitation, left border, and center value of intuitionistic triangular fuzzy numbers to get better fuzzy numbers than classical ones. Assessing athlete exhaustion demands this precision, since even little data changes and gradients affect it. Equations (10) and (11) reveal that the FDSS-based athlete fatigue monitoring system describes fitness parameters using intuitionistic triangular fuzzy numbers

$$T_{Fatigue} = (Low, Moderate, High; 0.2) \quad (10)$$

$$T_{Sleep\ Quality} = (Poor, Fair, Good; 0.1). \quad (11)$$

These representations allow FDSS to accommodate uncertainty in athlete-related data by considering a value's fuzzy set membership and hesitation degree. FDSS helps assess athletes' weariness using intuitionistic triangular fuzzy numbers, which better convey ambiguity. These easily understood representations allow the decision-support system to gracefully navigate athlete weariness in sports science's ever-changing and complex realm.

3.4 Real-Time Monitoring Paradigm

Theoretically, the framework presents a new way of monitoring athlete fatigue, prioritizing real-time observations above all else. The foundation of this conceptual growth is realizing that athlete tiredness is dynamic and ever-changing. Equation (12) shows the paradigm shift toward real-time monitoring, which is represented as $R(P)$, where P stands for physiological and performance data, and R symbolizes the real-time component

$$R(P) \rightarrow Real - Time\ Monitoring\ Paradigm. \quad (12)$$

Adhering to this paradigm, the FDSS-RAFM model is painstakingly designed to surpass traditional monitoring methods. The model captures the concept of real-time monitoring by giving immediate evaluations according to the present values of the physiological (P_t) and performance (Q_t) variables, as shown in Eq. (13)

$$FDSS - RAFM\ Real - Time\ Operation = S(P_t, Q_t). \quad (13)$$

Incorporating real-time physiological and performance factors, the variable S captures the model's functioning, allowing for accurate and rapid fatigue assessments. The move to real-time surveillance has serious, attainable repercussions. These rapid evaluations help athletes, coaches, and other sports professionals make better decisions

$$Decision\ Support = T(S(P_t, Q_t)) \quad (14)$$

Equation (14), utilizing real-time evaluations, incorporates a decision-support function (T) to minimize fatigue-induced performance reductions instantly. FDSS-RAFM uses proactive real-time monitoring. Its purpose is to ensure that decision-making and evaluation tools work rapidly to alleviate fatigue-related performance decreases

$$FDSS - RAFM\ Impact = U(Dt, S(P_t, Q_t)). \quad (15)$$

Equation (15) shows that the model immediately mitigates fatigue-related decreases. U reflects the impact, considering real-time evaluations and athlete dynamics (Pb). Due to its novel methodologies, this model is a game-changer for implementing the theoretical framework's real-time monitoring paradigm. It reduces the influence of weariness on performance by giving accurate and quick evaluations and judgments during training and contests.

3.5 Adaptability and Refinement Through Fuzzy Inference

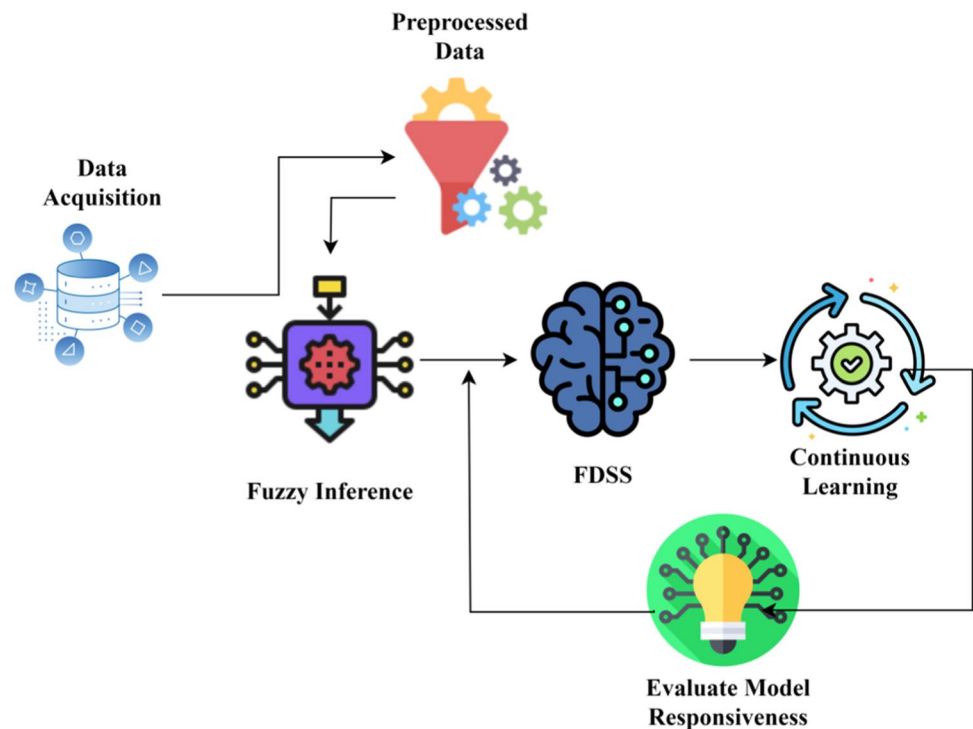
Fuzzy inference permits complex adaptation and refinement, essential to the FDSS-RAFM model (Fig. 3). The model may change and improve its real-time decision-making process to respond to changing inputs, fostering continual learning.

The concept from the Fuzzy inference theory shows the model's complex judging ability. Equation (16) represents fuzzy inference on pre-processed input D as $I(D)$

$$I(D) \rightarrow Adaptive\ Decision - Making. \quad (16)$$

Wearable accelerometers, heart rates, and perspiration sensors helped athletes track their performance throughout training and competition. These sensors monitored movement, core temperature, fluid levels, and HRV in real time. Raw data were pre-processed to reduce noise and signal distortions. The noise was filtered

Fig. 3 Adaptability and refinement through fuzzy inference



with a low-pass filter and standardized data to ensure device consistency. This study filled in missing values by interpolation to complete the dataset. We tested the system's accuracy and dependability in actual and virtual environments. Established criteria defined athletes as exhausted or not. The fuzzy decision-support system combined these metrics to evaluate real-time fatigue. This methodical process ensured high-quality and reliable data for model training and testing. The FDSS-RAFM model adds fuzzy sets and rules to pre-processed data for ongoing learning. The model's ability to change fuzzy sets and rules shows its versatility. This trait is continuous learning (L)

$$FDSS - RAFM \text{ Operation} = J(D, L). \quad (17)$$

Integrating data processing and continual learning ensures the model's overall functioning (J in Eq. (17) is adaptable and refined). Fuzzy inference lets the model adapt to sporting scenarios. This flexibility is needed to track an athlete's tiredness, because performance and physiological variables change

$$FDSS - RAFM \text{ Responsiveness} = K(P_t, Q_t, L_t). \quad (18)$$

In Eq. (18), K represents the model's responsiveness by integrating (P_t) , (Q_t) , and continuous learning. This improves model decision-making over time. The FDSS-RAFM model improves decision-making and adaptability to diverse sports via fuzzy inference. Because sports events are dynamic, this model's adaptability improves real-time athlete tiredness measurement.

Smartwatches and biosensors can measure temperature, blood oxygen levels, heart rate, and movement in real time as part of the physiological signal collection. This study used the z-score to identify data points with a dispersion greater than three standard deviations from the mean. A failing sensor or unusual physiological reactions like fast position or signal strength changes may be causes. IQR was also utilized to identify outliers outside the 25th to 75th percentile range. This statistical strategy may remove data errors from unexpected motions, malfunctioning sensors, or external interferences.

3.6 Integration of Stakeholder Feedback

For the FDSS-RAFM model to succeed in real-world sports and inspire cooperation, stakeholder feedback must be incorporated into the theoretical framework. Equation (19) uses I for integration and F for coach, sports expert, and player input. $I(F)$ shows this is conceptual

$$I(F) \rightarrow \text{Integration of Stakeholder Feedback.} \quad (19)$$

Feedback loops aid in monitoring teamwork in the FDSS-RAFM paradigm. Equation (20) shows the collaborative technique as $M(F_t)$, where M is the model operation that considers feedback at time b

$$\text{FDSS} - \text{RAFM Model Operation} = M(F_t). \quad (20)$$

For the F_t , the method ensures that stakeholder feedback contributes to constantly refining and enhancing the model's performance. This integration may improve stakeholder collaboration. The model is dynamic and responds to input, so coaches, sports experts, and athletes must monitor it

$$\text{Collaborative System} = C(I(F_t), M(F_t)). \quad (21)$$

The variable C in Eq. (21), representing the collaborative system, considers both initial stakeholder feedback integration ($I(F_t)$) and ongoing model operation with feedback. Since stakeholder feedback is dynamic in sports, the FDSS-RAFM system is designed to succeed and adapt. The model can be easily adapted, because athletes, coaches, and sports experts have different needs and perspectives

$$\text{Relevance and Effectiveness} = R(E, I(F_t), M(F_t)). \quad (22)$$

The variable R in Eq. (22) represents utility and significance. This metric considers evolving needs (E) and stakeholder feedback integration ($I(F_t)$), $M(F_t)$). Including stakeholder feedback in the monitoring process can make the FDSS-RAFM model collaborative and dynamic. Integrating the model allows it to be extended, altered, and maintained to be relevant in the dynamic and practical world of sports.

3.7 Longitudinal Analysis and Continuous Learning

The theoretical framework that ensures efficacy relies on robust continuous learning and longitudinal analysis. LAC, which stands for longitudinal analysis and continuous learning, depicts the model's ongoing progress. As illustrated in Eq. (23), the FDSS-RAFM model iteratively refines its fuzzy logic rules to learn the concept, improving its accuracy

$$P_t = \text{FDSS} - \text{RAFM Model Learning at Timet.} \quad (23)$$

Combining historical data from the Database Management System (DMS) for longitudinal analysis allows the model to learn from past trends and adapt to new sporting events (Eq. 24)

$$DMS \xrightarrow{\text{Historical Data}} LAC. \quad (24)$$

In Eq. (25), the equation LAC_t for time t represents the iterative learning process, continuously refining the model's predictive capabilities

$$LAC_t = \text{Longitudinal Analysis and Continuous Learning at Timet.} \quad (25)$$

This strategy improves the model's long-term forecast by accounting for sports' ever-changing nature. Equation (26) demonstrates that combining longitudinal analysis and continuous learning improves the FDSS-RAFM model's adaptive ability (E). This will enable the model to handle athlete fatigue's complexity

$$A = LAC_r. \quad (26)$$

Long-term model effectiveness depends on its ability to learn and adapt to athlete situations. The theoretical framework uses an adjustable, learning-focused FDSS-RAFM model. The model evolves, improves, and maintains efficacy through continual learning and longitudinal analysis, not abstract notions. It prepares for accurate and fast tiredness assessments in different athletic settings.

Fuzzy Decision Support Systems analyze athlete tiredness using fuzzy logic, a revolutionary sports science technology. It challenges easy categorization by considering the dynamic relationship between performance and physiological characteristics. FDSS-RAFM includes fuzzy rule sets, intuitionistic triangle fuzzy numbers, and membership functions. Fuzzy inference enhances R(P), a real-time monitoring paradigm. Priorities include collaboration and stakeholder input. Latent Dirichlet Allocation (LAC) stresses continuous learning and longitudinal analysis to maintain model efficacy. These sections explain the FDSS-RAFM model's adaptability, sophistication, and real-time tiredness evaluation.

4 Experimental Analysis and Discussion

4.1 Experimental Setup

The Fuzzy Decision Support Systems for Real-Time Athlete Fatigue Monitoring model was tested in a simulated sports setting. Given the broad demographic of athletes in different sports, a big dataset was available [26]. Real-time performance and physiological data can track athletes' training and sports. Modern sensors monitor heart rate, temperature, and respiration. Tracking devices measured speed, reaction time, and dexterity. Data were cleansed, filtered, and standardized during pre-processing for accuracy and consistency.

4.2 Performance Metrics and Evaluation

This study uses several indicators to evaluate the model's real-time fatigue performance. Depending on exhaustion, circumstances, and training intensity, it considers it. A battery of testing ensures model accuracy. Sensitivity, specificity, accuracy, DAE, and RTE are examples of this analysis. The FDSS-RAFM model's sophisticated functioning shows its advantages over fuzzy-based models and suggests that it could revolutionize player exhaustion monitoring during intensive athletic events.

4.2.1 Sensitivity Analysis

The fuzzy decision-support system FDSS-RAFM model consistently detected fatigued athletes across training intensities when tested against competing methods. The model's exceptional sensitivity at extremely low training intensities (0.98), demonstrated in Fig. 4, and strong sensitivity (0.97) even at high training intensities are confirmed. Amazingly, it surpasses the Triangular Fuzzy Inference Model (TFIM), Extended Fuzzy Logic Model (EFLM), and Fuzzy C-Means (FCM) at every energy level. Though less sensitive than lower intensities, the FDSS-RAFM model trumps FCM, TFIM, and EFLM. The model's high sensitivity indicates that it can detect fatigue quickly, allowing coaches to adjust training programs proactively. Although their sensitivity decreases gradually with increasing training intensity, FDSS-RAFM repeatedly proves superior to TFIM, EFLM, and FCM by performing better on every level. Coaches and sports professionals can rely on the model to detect fatigue quickly, allowing them to adjust training loads promptly. The findings provide more

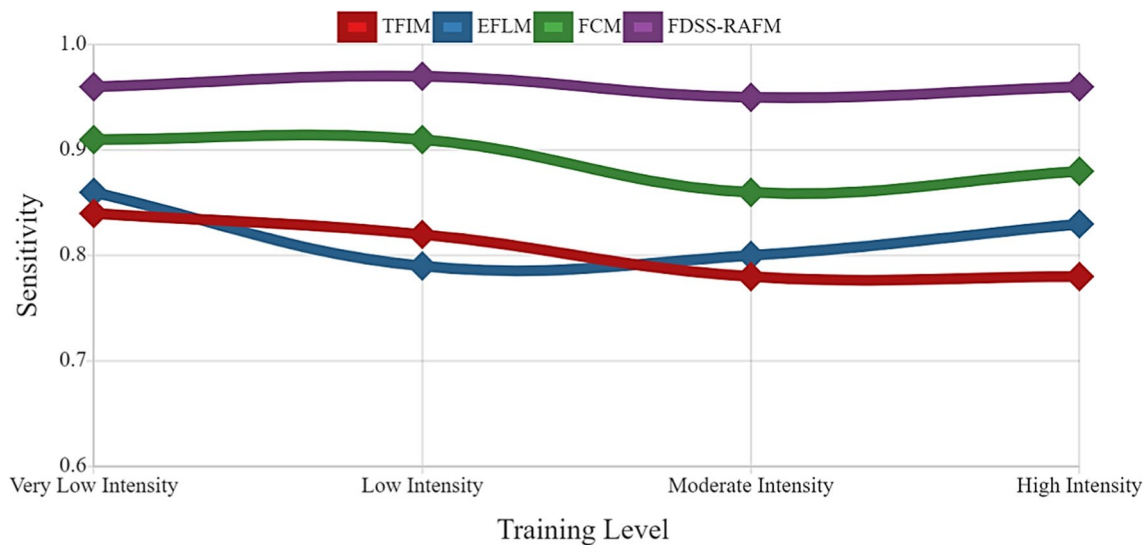


Fig. 4 Sensitivity analysis of FDSS-RAFM and other models with training intensity

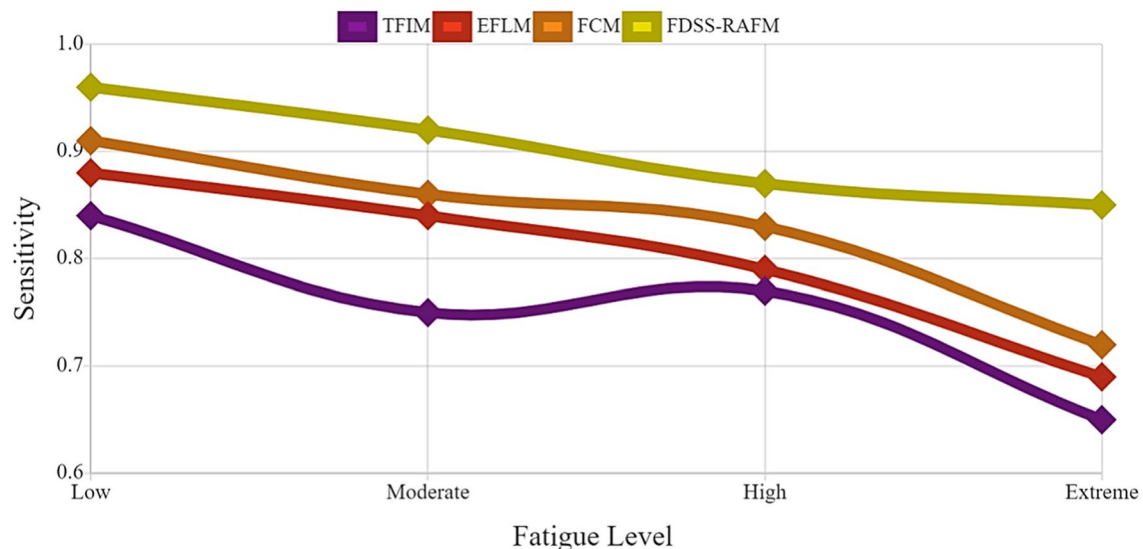


Fig. 5 Sensitivity analysis of FDSS-RAFM and other models with fatigue level

evidence that fuzzy decision-support systems, particularly the FDSS-RAFM model, can improve athletes' well-being regardless of the intensity of their training.

Among the many fatigue levels that athletes may experience, the FDSS-RAFM model is one of the most sensitive. Figure 5 shows that it is most sensitive at low-fatigue levels (0.96), suggesting that it can detect even faint indications of exhaustion. The model's consistent performance is demonstrated by its high sensitivity even at moderate fatigue levels (0.92). The model shows a strong sensitivity of 0.87 even when fatigue levels are high, indicating that it can still accurately detect fatigue even when its intensity is high. The model's sensitivity of 0.85 remains commendable at extreme fatigue levels, offering valuable insights into athletes going through extreme fatigue. The FDSS-RAFM model's sensitivity is high across all fatigue levels, demonstrating its reliability in detecting intensity fatigue. It provides accurate and detailed information about athletes' fatigue states, which coaches and sports professionals can rely on to create individualized interventions. Some

Fig. 6 Specificity analysis of FDSS-RAFM and other models with training intensity

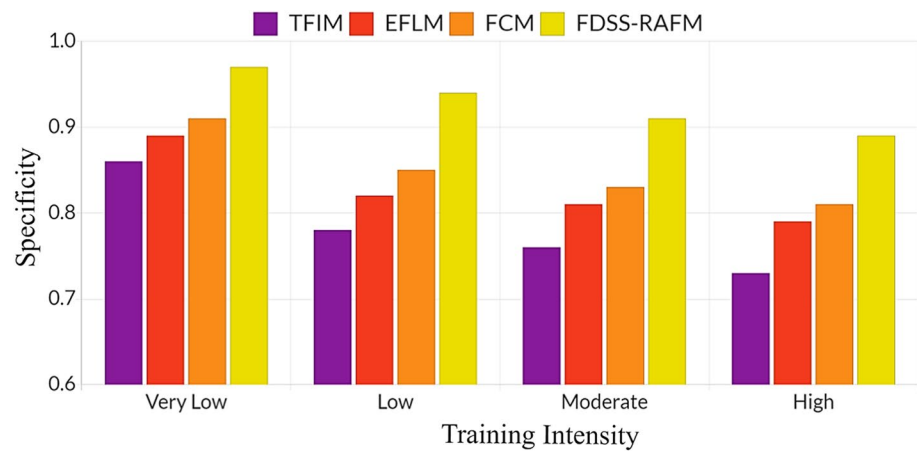
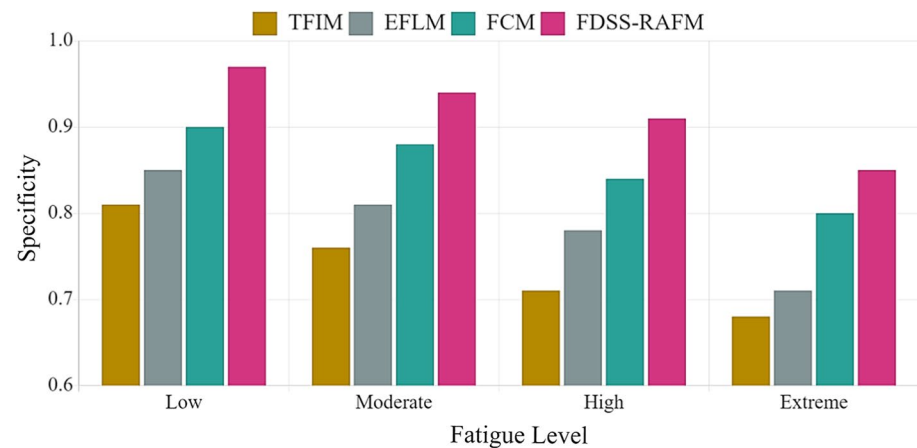


Fig. 7 Specificity analysis of FDSS-RAFM and other models with fatigue level



models, such as TFIM and EFLM, exhibit a loss of sensitivity with increasing fatigue levels, indicating that they may be unable to detect fatigue under more stressful situations. All fatigue levels show that the FDSS-RAFM model is more robust than FCM.

4.2.2 Specificity Analysis

Athletes who are not fatigued or only mildly fatigued can be easily identified using the FDSS-RAFM model. Figure 6 shows that it continuously surpasses other fuzzy-based models, especially at low and very low training intensities. The model's capacity to distinguish between athletes who are not fatigued even under ordinary training circumstances is demonstrated by its high specificity even at low training intensities. The model's effectiveness in identifying non-fatigued athletes even when elevated training stress is indicated by its commendable specificity (0.89), maintained even under high training intensity. Reliability in identifying non-fatigued or low-fatigue athletes is emphasized by the high specificity of the FDSS-RAFM model across all training intensities. The model can be relied on by coaches and sports professionals to accurately identify athletes who are not fatigued, which will help develop individualized training programs. Regarding comparative model performance, FDSS-RAFM always beats TFIM, EFLM, and FCM models, especially when training intensities are low or extremely low. Coaches who want to modify training interventions depending on precise evaluations of athletes' exhaustion levels will find the FDSS-RAFM model invaluable due to its high specificity.

Athletes at various levels of exhaustion can be easily identified using the FDSS-RAFM model. Its precision across all tiredness levels, from mild to severe, proves its reliability in detecting rested athletes. Figure 7 shows that the model outperforms the reputable TFIM and EFLM in weariness. Regarding specificity, FDSS-RAFM

Fig. 8 Accuracy rate of FDSS-RAFM and other models with training intensity

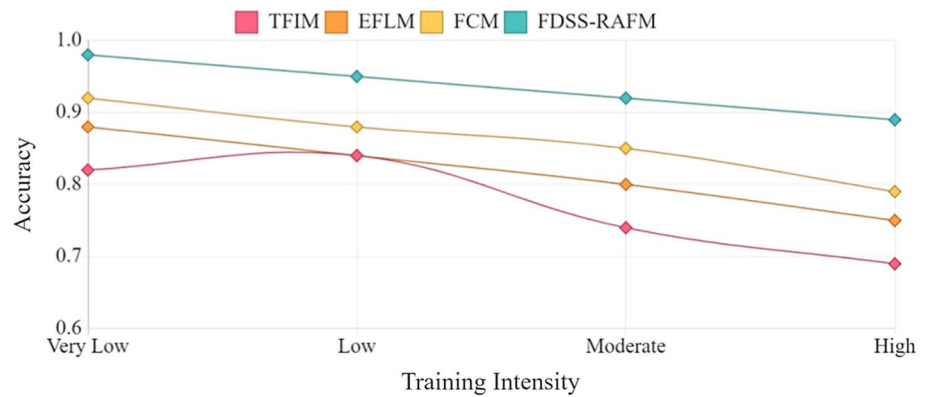
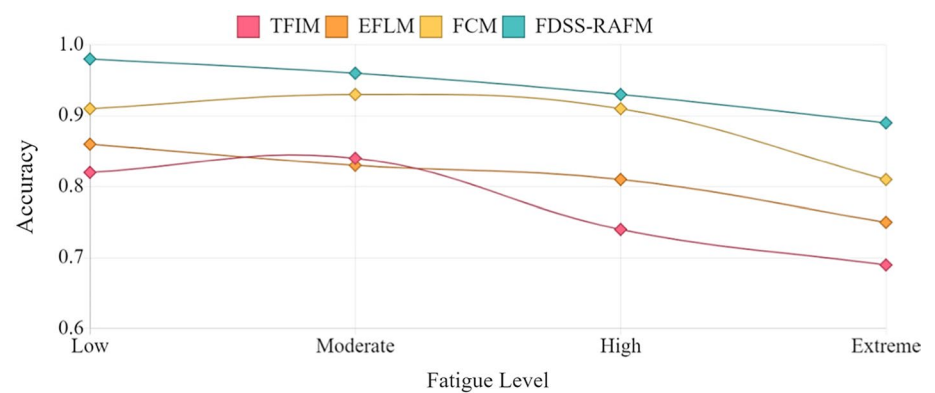


Fig. 9 Accuracy rate of FDSS-RAFM and other models with fatigue level

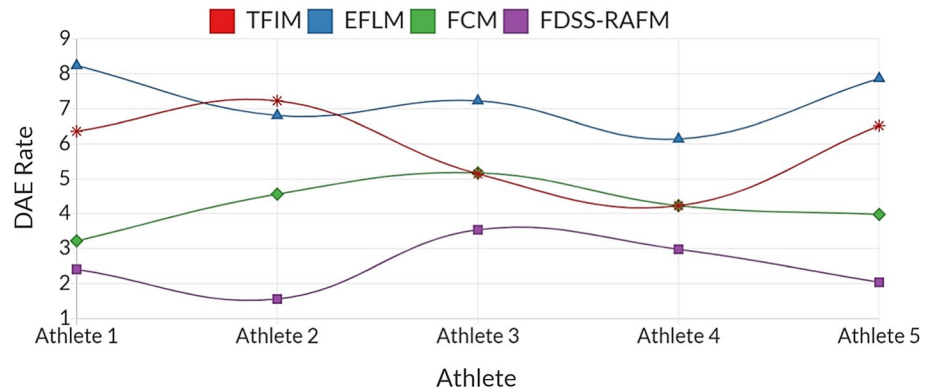
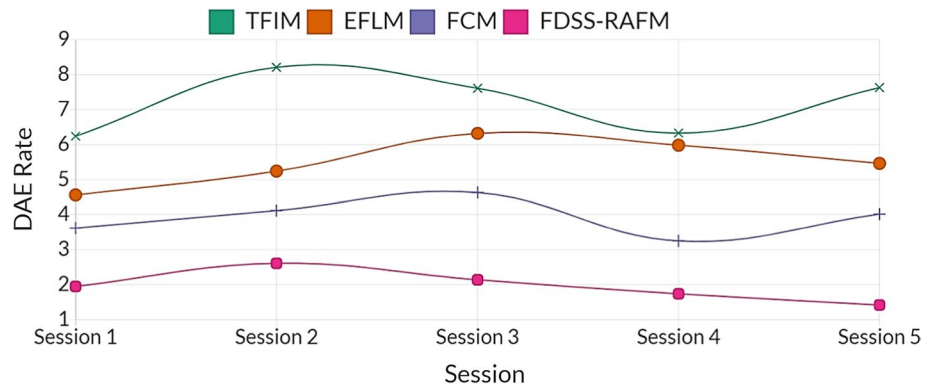


often exceeds FCM in distinguishing between well-rested athletes. The FDSS-RAFM model classifies athletes by exhaustion level to help coaches target interventions. Despite their considerable specificity, FDSS-RAFM outperforms TFIM and EFLM at all fatigue levels. Coaches can customize therapies to player weariness using the FDSS-RAFM model. No matter the athlete's exhaustion, the model will show they are not tired.

4.2.3 Accuracy Analysis

The fuzzy-based FDSS-RAFM model was compared to models with different training intensities. Figure 8 indicates that the model can detect weariness with minimal training. Its durability is shown by its 0.95 tiredness prediction accuracy when trained at modest intensities. It exceeds expectations with an accuracy rate 0.92 at a typical training effort. With an accuracy rate of 0.89, it can accurately predict weariness during intense training sessions. TFIM, EFLM, and FCM perform well at varying training intensities; however, FDSS-RAFM always outperforms them. Coaches and players benefit from the model's high accuracy rates in real-time fatigue assessments during rigorous workouts. The results highlight how the FDSS-RAFM model can be adjusted to different training intensities, which makes it a helpful tool for monitoring athlete fatigue in various sports. The high accuracy rates add credibility and dependability, making it more suitable for real-world, ever-changing sporting settings.

The fuzzy-based FDSS-RAFM model has been tested against other models at varying degrees of exhaustion. As shown in Fig. 9, the model achieves an outstanding accuracy rate of 0.98 under conditions of low fatigue and maintains a high accuracy rate of 0.96 under conditions of moderate fatigue. Additionally, it successfully detects increasing fatigue levels, as evidenced by its commendable accuracy rate of 0.93, even at high fatigue. The model successfully handles and predicts extreme fatigue states with a notable accuracy rate of 0.86, even in situations with extreme fatigue. Despite TFIM, EFLM, and FCM demonstrating passable accuracy rates across different fatigue levels, the FDSS-RAFM model perpetually surpasses them. The model is an influential and trustworthy

Fig. 10 DAE rate of FDSS-RAFM and other models with five athletes**Fig. 11** DAE rate of FDSS-RAFM and other models with five sessions

resource for sports professionals and coaches due to its high accuracy rates; it helps them make informed decisions about training programs for athletes depending on their fatigue levels. The FDSS-RAFM model can accurately assess fatigue in real time, which bodes well for sports science. Its durability and accuracy make it perfect for high-intensity sports where seconds matter.

4.2.4 Dynamic Adaptation Error Analysis

Figure 10 compares the FDSS-RAFM model to other DAE rate models using five athletes. To consistently outperform competing models shows that FDSS-RAFM can adapt to varied sporting scenarios. Athlete 1's FDSS-RAFM was most flexible to fatigue levels and had the highest DAE rate (2.41). With Athlete 2, FDSS-RAFM demonstrated the best adaptability to changing conditions with the lowest DAE rate (1.56). Effective adaptation was suggested by FDSS-RAFM's competitive DAE rate (3.54), which it maintained for Athlete 3. An advantageous DAE rate (2.98) for Athlete 4 was observed in FDSS-RAFM, suggesting effective adaptation. The DAE rate for Athlete 5 was 2.04, the lowest for FDSS-RAFM. All athletes significantly improved DAE rates, with FDSS-RAFM consistently outperforming TFIM and EFLM models. FDSS-RAFM demonstrated superior adaptability across varying degrees of fatigue, surpassing FCM in DAE rates for all athletes. The model seems good at adjusting its predictions to changes in athlete fatigue, which would explain the low DAE rates. This means that evaluations are accurate and can be done quickly. It shows its potential for real-time monitoring of athlete fatigue across different athletes.

Five separate sessions were used to analyze the FDSS-RAFM model's DAE rates. In comparison to TFIM (6.24), EFLM (4.56), and FCM (3.62), FDSS-RAFM showed the lowest DAE rate (1.95), as shown in Fig. 11. Based on these results, it seems that Session 1 is the best time for FDSS-RAFM to adjust to variations in weariness. Session 2 results show that FDSS-RAFM is more adaptable to changing conditions than competing models, consistently outperforming them with the lowest DAE rate (2.61). FDSS-RAFM's ability to adapt to changes in

Fig. 12 RTE rate of FDSS-RAFM and other models with five athletes

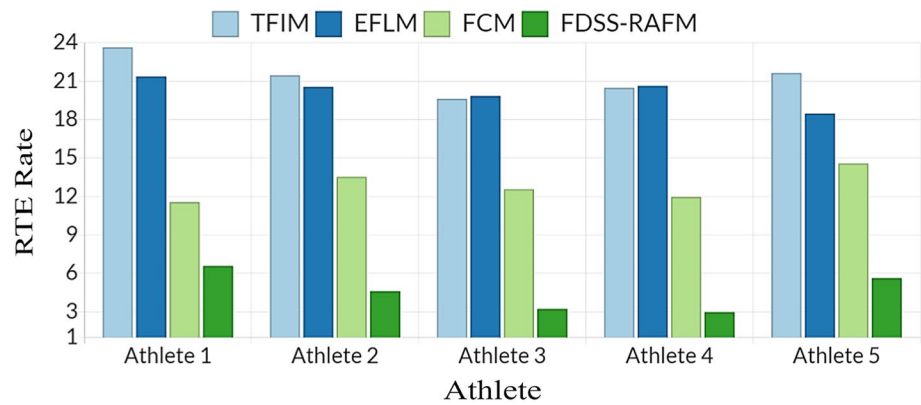
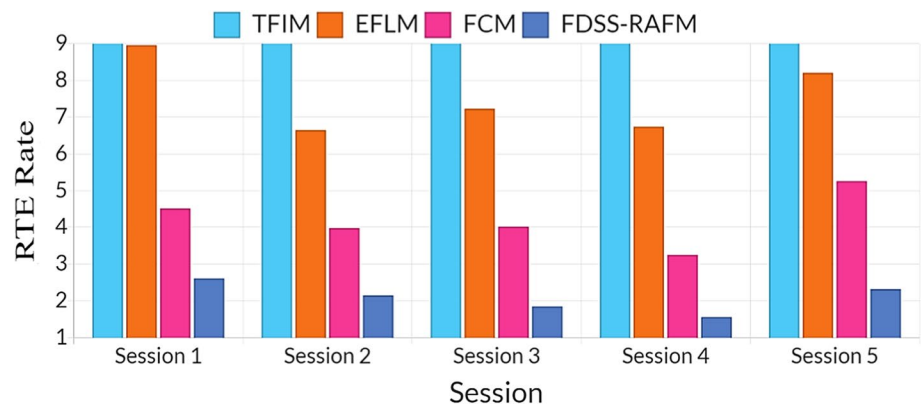


Fig. 13 RTE rate of FDSS-RAFM and other models with five sessions



fatigue is demonstrated by its competitive DAE rate (2.14), which it maintains throughout Session 3, a favorable DAE rate (1.74), and the lowest DAE rate (1.42), which it records during Session 5. The results reveal that FDSS-RAFM is better at responding to training fatigue than TFIM, EFLM, and FCM. This benefit is constant throughout sessions. The low DAE rates for FDSS-RAFM indicate that the model properly adjusts for athlete tiredness between sessions. Sports industry professionals and coaches can use FDSS-RAFM to make session-specific decisions to improve athlete health and performance. The DAE study concludes that FDSS-RAFM is a good instrument for measuring athletes' current weariness. It always handles training environment changes better than other fuzzy-based models.

4.2.5 Response Time Error Analysis

Figure 12 shows that the FDSS-RAFM model has a better response time error rate when dealing with athlete data. Greater RTE in the TFIM model indicates inaccurate and delayed response times. Despite a decrease in RTE, the EFLM model still displays significant delays, indicating potential enhancement. Even though the FCM model has a lower RTE, it is still behind the FDSS-RAFM. Among all athletes, the FDSS-RAFM model has the lowest RTE, demonstrating its consistent superiority over other models. It shows that the model works well for different athletes, since there is a constant improvement in response time efficiency. It highlights the efficiency of FDSS-RAFM in making decisions in real time, as it outperforms TFIM, EFLM, and FCM. Personalized and efficient responses are ensured through the athlete-specific analysis, which helps to tailor the model to individual needs; enhanced operational efficiency results from lower RTE, which means faster decision-making in training or competitive scenarios. The model's effectiveness, particularly in real-time situations, makes it an attractive tool for sports professionals and coaches who want to make quick decisions based on evaluations of athletes' exhaustion.

Figure 13 shows that the FDSS-RAFM model is more efficient based on Response Time Error rate analysis than other fuzzy-based models. Inefficiencies in adapting to various session-specific conditions are indicated by TFIM performance, which shows higher RTE. The RTE displayed by EFLM performance varies across sessions, with occasional spikes. While FCM's performance is generally good, with lower RTE in most sessions, it falls short of FDSS-RAFM, suggesting room for improvement. FDSS-RAFM's consistent outperformance and lowest RTE across all sessions illustrate its adaptability. It is reliable for real-time session judgments due to its stable operation and low RTE. The proposed FDSS-RAFM model is ideal for training and competition due to its low RTE across sessions.

Several indicators show that the Fuzzy Decision Support System-integrated real-time athlete fatigue monitoring model performs well. No matter the training intensity or lowness, the FDSS-RAFM model outperforms the competition in weariness detection. These include TFIM, EFLM, and FCM. Despite exhaustion, this sensitivity allows for quick plan modifications. The program is highly precise across training intensity and fatigue, so it always finds rested athletes. Coaches desire accurate fatigue measurements to improve training efficacy. The model consistently outperforms TFIM, EFLM, and FCM in real-time fatigue assessments. This model consistently outperforms the competition in DAE rates, demonstrating its adaptability to athlete situations. The model's RTE analysis consistently provides the lowest RTE rates across athletes and sessions, demonstrating its fast response to athlete data. The FDSS-RAFM model uses fuzzy logic to provide athletes with specific information on exhaustion phases to help them stay healthy and perform well in various sports.

Data privacy and transparency are athlete monitoring systems' biggest ethical concerns. These systems manage sensitive personal data, including health measurements and performance statistics; hence, GDPR is needed to protect rights. Protecting sensitive data requires encryption, secure access, and clear data ownership policies. Decision-making must be transparent to build stakeholder trust. This study uses XAI models to ensure doctors, coaches, and players understand how data are utilized to make decisions. Platforms without moral standards risk losing users and face criticism from individuals concerned about overreach, abuse, and data breaches.

Compliance with the General Data Protection Regulation (GDPR) emphasizes consent-based data collecting and user rights, including portability and deletion, to improve the system and ensure ethical usage of sensitive athlete data. Explainable AI (XAI) models may help players, coaches, and medical professionals collaborate and feel more comfortable during a monitoring program. Although real-time feedback system recommendations cannot be articulated, they provide significant health information. Anti-data abuse solutions include advanced data storage, transport encryption, and multi-factor authentication for system access. These measures prevent unauthorized access and ensure that data are used only for health outcomes and efficiency.

5 Conclusion

Using the FDSS-RAFM model, this study would address athlete weariness in sports science in a novel way. Fuzzy logic is used to modify the paradigm, since human performance data are imprecise and uncertain. Traditional monitoring methods fail to capture weariness's complexity and changeability. FDSS-RAFM has revolutionized sports science by providing professionals, coaches, and athletes with a cutting-edge resource. It may improve real-time performance, injury prevention, and training regimens. The study emphasizes developing new, specific techniques to reduce athlete tiredness quickly. We introduce FDSS-RAFM, a fuzzy-based decision-support system that is smarter and more situationally aware. This strategy reduces weariness-related performance declines in competitive sports by giving coaches more data during practice and games. The model's capacity to detect weariness at varying intensities and levels shows fuzzy logic's real-time accuracy. The FDSS-RAFM paradigm calls for widespread implementation of cutting-edge technology in athletics to usher in a new era of athlete health and performance. The FDSS-RAFM approach is better, but cameras and wearables raise ethical and privacy problems. Possible factors influencing its efficacy include data transmission delays and the unique traits of each athlete. If the model is to be widely used in sports science, future research should concentrate on creating instructional

programs for coaches and sports professionals to help them better understand and use it, as well as to solve privacy concerns and add more physiological and performance data. The experimental results show that the proposed FDSS-RAFM model increases the sensitivity ratio by 97%, specificity ratio by 89%, accuracy ratio by 96%, and dynamic adaptation error analysis of 2.41 compared to other existing models. Wearable sensors must be accurate and reliable to avoid data quality issues caused by environmental interferences or malfunctions, which is one of the numerous challenges to their widespread use. Because of the need for a reliable, low-latency infrastructure for real-time analysis of multi-modal data, another concern is the computational efficiency of the system.

Author Contributions The author confirms being the sole contributor of this work and has approved it for publication.

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Data Availability The data used to support the findings of this study are all in the manuscript.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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